CHAPTER 02: STRUCTURE OF ATOM

Comprehensive Theory, Reaction Mechanisms & Formula Roadmap (Class 11 & 12)

1. HISTORICAL EVOLUTION OF ATOMIC STRUCTURE

Daltons Atomic Theory (1808): Matter consists of indivisible atoms.

Discovery of Subatomic Particles proved atom is divisible into electrons, protons, and neutrons.

Cathode Ray Discharge Tube Experiments (J.J. Thomson, 1897): Discovered negatively charged electrons.

Charge-to-mass ratio of electron: e/m = 1.758820 x 10^11 C/kg.

Millikans Oil Drop Experiment (1909): Determined fundamental electronic charge e = -1.6022 x 10^-19 C.

Mass of electron: m_e = e / (e/m) = 9.1094 x 10^-31 kg.

2. ATOMIC MODELS & NUCLEAR DISCOVERY

Thomsons Model of Atom (Plum Pudding Model): Uniform sphere of positive charge with electrons embedded.

Rutherfords Alpha-Particle Scattering Experiment (1911): Gold foil bombarded with fast alpha particles.

Key Observation: 99.9% particles passed undeflected; tiny fraction deflected by large angles (>90 degrees).

Conclusion: Atom is mostly empty space; positive charge and mass concentrated in tiny central nucleus.

Nuclear radius ~ 10^-15 m compared to atomic radius ~ 10^-10 m.

Drawback: Could not explain electronic stability or line spectra of hydrogen.

CHAPTER 02: STRUCTURE OF ATOM (PART 2)

Wave Nature of Light, Photoelectric Effect & Bohrs Postulates

3. ELECTROMAGNETIC RADIATION & PLANCK S QUANTUM THEORY

Wave Nature: c = nu x lambda (c = 3.0 x 10^8 m/s in vacuum). Wave number nu_bar = 1 / lambda.

Plancks Quantum Hypothesis (1900): Radiant energy emitted or absorbed in discrete packets called quanta.

Energy of quantum: E = h x nu = (h x c) / lambda, where Planck constant h = 6.626 x 10^-34 J s.

Photoelectric Effect (Hertz, 1887 & Einstein, 1905): Emission of electrons from metal by light.

Kinetic energy of ejected photoelectron: K.E. = h x nu - h x nu_0 = h x nu - W_0 (Work function).

4. BOHR S MODEL FOR HYDROGEN ATOM (1913)

Postulate 1: Electrons move in fixed circular orbits called stationary states without radiating energy.

Postulate 2 (Quantization of Angular Momentum): m_e x v x r = n x (h / 2pi), where n = 1, 2, 3...

Postulate 3: Energy absorbed or emitted only during orbital transitions: Delta E = E2 - E1 = h x nu.

Radius of nth orbit: r_n = (n^2 x a_0) / Z, where Bohr radius a_0 = 52.9 pm (0.529 Angstrom).

Energy of nth orbit: E_n = -2.18 x 10^-18 x (Z^2 / n^2) J/atom = -13.6 x (Z^2 / n^2) eV.

CHAPTER 02: STRUCTURE OF ATOM (PART 3)

Hydrogen Spectral Series & Quantum Mechanics Foundations

5. HYDROGEN ATOMIC LINE SPECTRA

Rydberg Formula: nu_bar = 1 / lambda = R_H x Z^2 x [1/(n1^2) - 1/(n2^2)], R_H = 109,677 cm^-1.

Lyman Series (n1 = 1, n2 = 2, 3, 4...): Ultraviolet (UV) region.

Balmer Series (n1 = 2, n2 = 3, 4, 5...): Visible region of spectrum.

Paschen Series (n1 = 3, n2 = 4, 5...): Near-Infrared (IR) region.

Brackett Series (n1 = 4, n2 = 5, 6...): Infrared region.

Pfund Series (n1 = 5, n2 = 6, 7...): Far-Infrared region.

6. TOWARDS QUANTUM MECHANICS

Dual Behavior of Matter (de Broglie, 1924): lambda = h / p = h / (m x v).

Applicable to microscopic particles (electrons) with measurable wave properties.

Heisenberg Uncertainty Principle (1927): Delta x x Delta p >= h / (4pi).

It is impossible to measure simultaneously both exact position and momentum of a microscopic particle.

Significance: Ruled out concept of definitive circular Bohr orbits; established concept of probability.

CHAPTER 02: STRUCTURE OF ATOM (PART 4)

Quantum Numbers, Orbitals & Shapes

7. QUANTUM NUMBERS & ORBITAL ADDRESSES

Principal Quantum Number (n = 1, 2, 3...): Determines shell, size, and main orbital energy.

Azimuthal Quantum Number (l = 0 to n-1): Determines subshell and three-dimensional shape.

l = 0 (s, spherical); l = 1 (p, dumbbell); l = 2 (d, double dumbbell); l = 3 (f, complex).

Magnetic Quantum Number (m_l = -l to +l): Spatial orientation of orbital. Total orbitals = 2l + 1.

s has 1 orbital; p has 3 orbitals (px, py, pz); d has 5 orbitals (dxy, dyz, dzx, dx2-y2, dz2).

Spin Quantum Number (m_s = +1/2 or -1/2): Spin angular momentum of electron around its own axis.

8. NODAL PLANES & PROBABILITY DENSITIES

Radial Nodes = n - l - 1.

Angular Nodes = l.

Total Nodes = (n - l - 1) + l = n - 1.

Example 1: 3p orbital has n=3, l=1 -> Radial nodes = 3-1-1 = 1, Angular nodes = 1, Total = 2.

Example 2: 4d orbital has n=4, l=2 -> Radial nodes = 4-2-1 = 1, Angular nodes = 2, Total = 3.

CHAPTER 02: STRUCTURE OF ATOM (PART 5)

Rules for Filling Orbitals, Electronic Configuration & High-Yield DPP

9. RULES FOR FILLING ELECTRONS IN ORBITALS

Aufbau Principle: Orbitals filled in order of increasing energy: (n + l) rule.

Lower (n + l) orbital filled first. If (n + l) is identical, lower n orbital filled first.

Sequence: 1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d...

Paulis Exclusion Principle: No two electrons in an atom can have the same set of 4 quantum numbers.

Maximum 2 electrons per orbital, and they must have opposite spins (+1/2 and -1/2).

Hunds Rule of Maximum Multiplicity: Pairing in degenerate subshells begins only when each orbital is singly occupied.

Exceptional Stability: Half-filled (d5) and completely-filled (d10) subshells have extra symmetry and exchange energy.

Chromium (Z=24): [Ar] 3d5 4s1 (not 3d4 4s2); Copper (Z=29): [Ar] 3d10 4s1 (not 3d9 4s2).

10. HIGH-YIELD PRACTICE PROBLEMS & BOARD DERIVATIONS

Q1: Calculate wavelength of a photon emitted during transition from n=4 to n=2 in H-atom. (Ans: 486 nm)

Q2: Find the number of radial and angular nodes for 5d orbital. (Ans: Radial=2, Angular=2)

Q3: State why 4s orbital is filled before 3d orbital. Explain using (n+l) rule. (Ans: 4s has n+l=4, 3d has n+l=5)

Q4: Calculate de Broglie wavelength of an electron moving with 1% speed of light. (Ans: 2.42 x 10^-10 m)

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